CODE APEX

THE END IS COMING.

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Team Details

- Team name -
- Team leader Name -
- Problem Statement (Domain) -

Problem Statement & Target Need

- Who is affected
- The current gap
- Why it matters now

Proposed Solution & Innovation

- What makes it different
- Core mechanism
- Why it's feasible in this timeframe

System Architecture & Tech Stack

TECH STACK

- Frontend
- backend
- Database
- AI / APIs
- Hosting / DevOps

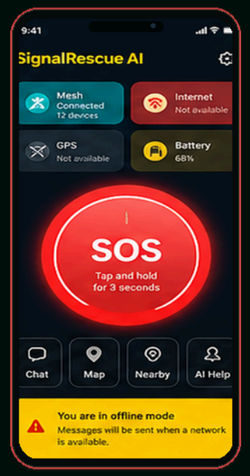
Key Features & UI/UX

CITIZEN & FIRST RESPONDER FIELD SUITE

[BLE 5.2 + WI-FI MESH]

RakshaNet Mobile (Android APK)

Zero-Internet Peer-to-Peer Mesh Network & Tactical SOS



SOS BEACON SCREEN



OFFLINE TOPO MAP

P2P Zero-Internet Mesh
BLE & Wi-Fi Direct store-and-forward relays packets across 15+ hops seamlessly.

One-Tap Instant SOS
3-second tactile hold triggers encrypted distress beacon with GPS & triage data.

Offline Tactical Maps
Cached OpenStreetMap vector tiles & peer proximity circles with zero cellular coverage.

On-Device Edge AI
Offline intelligence engine for medical triage, survival guidance & resource allocation.

TACTICAL DISASTER COMMAND & AGENCY HQ

[DUAL DB: SQLITE + MONGO]

RakshaNet Command Center

Real-Time Tactical GIS, AI Triage & Fleet Optimization



OVERVIEW & TRIAGE DASHBOARD

DBSCAN Cluster Engine
Spatial AI clusters scattered SOS calls into prioritized incidents with casualty counts.

Tactical GIS & Black Holes
Tracks mesh relays, mobile nodes & detects telemetry black holes (>75% packet loss).

MILP Fleet Optimization
Mathematical solver optimizes dispatch for NDRF boats, 4WD vehicles & paramedics.

CAP v1.2 & CAD Feeds
Standardized XML & JSON feeds for NDMA, SDRF & emergency response interoperability.

Implementation Roadmap & Scope

What's built today, what's in progress, and what's planned beyond the hackathon

Expected Impact & Scalability

Who benefits, how big the opportunity is, and how this grows beyond a demo.

VISHWAKARMA INSTITUTES VI

AISF
ARTIFICIAL INTELLIGENCE STUDENT FORUM

VCA

CODE APEX

BUILD BEFORE IT DOES.

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THANK YOU